

Operating Systems and System Programming

Skill-3

Task-1:

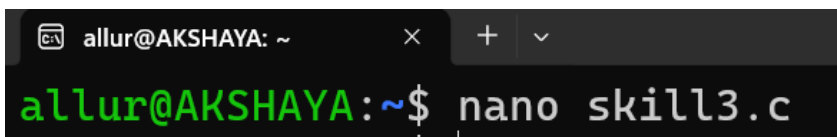
Aim

To implement command history and navigation using escape sequences, and to manage dynamically allocated buffers and linked lists while preventing buffer overflow and memory leaks.

Algorithm

1. Start the program.
2. Read and store user commands in the input buffer.
3. Use Up/Down arrow escape sequences to navigate through command history.
4. Dynamically allocate memory for storing commands.
5. Resize the buffer when it becomes full.
6. Store commands using a linked list.
7. Display the stored commands.
8. Free all allocated memory.
9. Verify memory usage using Valgrind.
10. Stop.

Step-1:



```
allur@AKSHAYA: ~$ nano skill3.c
```

Step-2:

```
GNU nano 8.7.1 skill3.c *
#include <stdio.h>
#include <string.h>
#include <termios.h>
#include <unistd.h>

#define MAX 5
#define SIZE 100

int main()
{
    char history[MAX][SIZE], input[SIZE], ch;
    int count = 0, pos, hpos;
    struct termios old, new;

    tcgetattr(0, &old);
    new = old;
    new.c_lflag &= ~(ICANON | ECHO);
    tcsetattr(0, TCSANOW, &new);

    while (1)
    {
        printf("\nmysHELL> ");
        fflush(stdout);
```

```
pos = 0;
hpos = count;

while (1)
{
    ch = getchar();

    if (ch == '\n')
    {
        input[pos] = '\0';
        break;
    }

    if (ch == 27) // Escape sequence
    {
        getchar(); // [
        ch = getchar();

        if (ch == 'A' && hpos > 0) // Up
            hpos--;
        else if (ch == 'B' && hpos < count) // Down
            hpos++;
    }
}
```

```
allur@AKSHAYA: ~  
GNU nano 8.7.1 skill3.c *  
    if (ch == 'A' || ch == 'B')  
    {  
        if (hpos < count)  
            strcpy(input, history[hpos]);  
        else  
            input[0] = '\0';  
  
        pos = strlen(input);  
        printf("\r\033[kmyshell> %s", input);  
        fflush(stdout);  
    }  
else if (ch == 127 && pos > 0) // Backspace  
{  
    pos--;  
    printf("\b \b");  
}  
else if (pos < SIZE - 1)  
{  
    input[pos++] = ch;  
    putchar(ch);  
}  
| }  
  
^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Exec  
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Just
```

```
allur@AKSHAYA: ~  
GNU nano 8.7.1 skill3.c *  
  
    if (strcmp(input, "exit") == 0)  
        break;  
  
    if (pos > 0)  
    {  
        if (count < MAX)  
            strcpy(history[count++], input);  
        else  
        {  
            for (int i = 1; i < MAX; i++)  
                strcpy(history[i - 1], history[i]);  
            strcpy(history[MAX - 1], input);  
        }  
    }  
  
    printf("\nYou entered: %s\n", input);  
}  
  
tcsetattr(0, TCSANOW, &old);  
return 0;  
}  
|  
  
^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Exec  
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Just
```

Step-3:

```
allur@AKSHAYA:~$ gcc skill3.c
allur@AKSHAYA:~$ ./a.out
```

```
myshell> hello
You entered: hello

myshell> This is a command
You entered: This is a command

myshell> |
```

Press the ↑ Up Arrow key once.

```
myshell> hello
You entered: hello

myshell> This is a command
You entered: This is a command

myshell> hello|
```

press ↓ Down Arrow

```
myshell> hello
You entered: hello

myshell> This is a command
You entered: This is a command

myshell> This is a command|
```

```
allur@AKSHAYA:~$ ./a.out

myshell> hello
You entered: hello

myshell> This is a command
You entered: This is a command

myshell> exitallur@AKSHAYA:~$ |
```

Task-2:

Step-1:

```
allur@AKSHAYA:~$ nano skill3-2.c
```

Step-2:

```
allur@AKSHAYA: ~ x + v
GNU nano 8.7.1 skill3-2.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

struct Node
{
    char *cmd;
    struct Node *next;
};

int main()
{
    char *buf = malloc(10);
    int size = 10, len;
    struct Node *head = NULL, *newnode, *temp;

    while (1)
    {
        len = 0;
        printf("myshell> ");

        while (1)
        {
```

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execut
^X Exit	^R Read File	^N Replace	^U Paste	^J Justif

```
allur@AKSHAYA: ~  
GNU nano 8.7.1 skill3-2.c *  
    char ch = getchar();  
  
    if (ch == '\n')  
        break;  
  
    if (len >= size - 1)  
    {  
        size *= 2;  
        buf = realloc(buf, size);  
    }  
  
    buf[len++] = ch;  
}  
  
buf[len] = '\0';  
  
if (strcmp(buf, "exit") == 0)  
    break;  
  
newnode = malloc(sizeof(struct Node));  
newnode->cmd = malloc(strlen(buf) + 1);  
strcpy(newnode->cmd, buf);  
newnode->next = head;  
|  
^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execu  
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justi
```

```
allur@AKSHAYA: ~  
GNU nano 8.7.1 skill3-2.c *  
    head = newnode;  
}  
  
printf("\nCommands:\n");  
temp = head;  
  
while (temp != NULL)  
{  
    printf("%s\n", temp->cmd);  
    temp = temp->next;  
}  
  
while (head != NULL)  
{  
    temp = head;  
    head = head->next;  
    free(temp->cmd);  
    free(temp);  
}  
  
free(buf);  
return 0;  
}|  
^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execu  
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justi
```

Step-3:

```
allur@AKSHAYA:~$ gcc skill3-2.c
allur@AKSHAYA:~$ ./a.out
myshell> hello
myshell> ls -l
myshell> This is a command
myshell> exit
```

```
Commands:
This is a command
ls -l
hello
```

```
allur@AKSHAYA:~$
```

Step-4: Install Valgrind

```
allur@AKSHAYA:~$ sudo apt update
[sudo: authenticate] Password:
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:2 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.7 kB]
Get:3 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.2 kB]
Hit:4 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:5 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Hit:6 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:7 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [492 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [97.8 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [243 kB]
Get:10 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [184 kB]
Fetched 1382 kB in 3s (527 kB/s)
17 packages can be upgraded. Run 'apt list --upgradable' to see them.
allur@AKSHAYA:~$
```

```
allur@AKSHAYA:~$ sudo apt install valgrind
Installing:
  valgrind

Installing dependencies:
  gdb          libdebuginfod-common  libipt2
  libbabeltrace1 libdebuginfod1t64  libsource-highlight-common
  libc6-dbg    libdw1t64             libsource-highlight4t64

Suggested packages:
  gdb-doc  gdbserver  valgrind-mpi  kcache-grind

Summary:
  Upgrading: 0, Installing: 10, Removing: 0, Not Upgrading: 17
  Download size: 27.1 MB
  Space needed: 111 MB / 1024 GB available

Continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 libdebuginfod-common all 0.194-4 [12.3 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 libdw1t64 amd64 0.194-4 [285 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute/main amd64 libbabeltrace1 amd64 1.5.11-5build1 [165 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute/main amd64 libdebuginfod1t64 amd64 0.194-4 [20.7 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute/main amd64 libipt2 amd64 2.1.2-3 [49.2 kB]
Get:6 http://archive.ubuntu.com/ubuntu resolute/main amd64 libsource-highlight-common all 3.1.9-4.3build2 [64.2 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute/main amd64 libsource-highlight4t64 amd64 3.1.9-4.3build2 [273 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute/main amd64 gdb amd64 17.1-2ubuntu1 [4173 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libc6-dbg amd64 2.43-2ubuntu2.3 [6407 kB]
Get:10 http://archive.ubuntu.com/ubuntu resolute/main amd64 valgrind amd64 1:3.26.0-0ubuntu1 [15.7 MB]
Fetched 27.1 MB in 3s (8216 kB/s)
Preconfiguring packages ...
Selecting previously unselected package libdebuginfod-common.
(Reading database ... 42283 files and directories currently installed.)
Preparing to unpack .../0-libdebuginfod-common_0.194-4_all.deb ...
Unpacking libdebuginfod-common (0.194-4) ...
Selecting previously unselected package libdw1t64:amd64.
Preparing to unpack .../1-libdw1t64_0.194-4_amd64.deb ...
Unpacking libdw1t64:amd64 (0.194-4) ...
Selecting previously unselected package libbabeltrace1:amd64.
Preparing to unpack .../2-libbabeltrace1_1.5.11-5build1_amd64.deb ...
Unpacking libbabeltrace1:amd64 (1.5.11-5build1) ...
Selecting previously unselected package libipt2:amd64.
Preparing to unpack .../3-libipt2_2.1.2-3_amd64.deb ...
Unpacking libipt2:amd64 (2.1.2-3) ...
Selecting previously unselected package libsource-highlight4t64:amd64.
Preparing to unpack .../4-libsource-highlight4t64_3.1.9-4.3build2_amd64.deb ...
Unpacking libsource-highlight4t64:amd64 (3.1.9-4.3build2) ...
Selecting previously unselected package gdb.
Preparing to unpack .../5-gdb_17.1-2ubuntu1_amd64.deb ...
Unpacking gdb (17.1-2ubuntu1) ...
Selecting previously unselected package libc6-dbg:amd64.
Preparing to unpack .../6-libc6-dbg_2.43-2ubuntu2.3_amd64.deb ...
Unpacking libc6-dbg:amd64 (2.43-2ubuntu2.3) ...
Selecting previously unselected package valgrind.
Preparing to unpack .../7-valgrind_1:3.26.0-0ubuntu1_amd64.deb ...
Unpacking valgrind (1:3.26.0-0ubuntu1) ...
Setting up libdebuginfod-common (0.194-4) ...
Setting up libdw1t64:amd64 (0.194-4) ...
Setting up libbabeltrace1:amd64 (1.5.11-5build1) ...
Setting up libipt2:amd64 (2.1.2-3) ...
Setting up libsource-highlight4t64:amd64 (3.1.9-4.3build2) ...
Setting up gdb (17.1-2ubuntu1) ...
Setting up libc6-dbg:amd64 (2.43-2ubuntu2.3) ...
Setting up valgrind (1:3.26.0-0ubuntu1) ...
```

```
Unpacking libdw1t64:amd64 (0.194-4) ...
Selecting previously unselected package libbabeltrace1:amd64.
Preparing to unpack .../2-libbabeltrace1_1.5.11-5build1_amd64.deb ...
Unpacking libbabeltrace1:amd64 (1.5.11-5build1) ...
Selecting previously unselected package libdebuginfod1t64:amd64.
Preparing to unpack .../3-libdebuginfod1t64_0.194-4_amd64.deb ...
Unpacking libdebuginfod1t64:amd64 (0.194-4) ...
Selecting previously unselected package libipt2.
Preparing to unpack .../4-libipt2_2.1.2-3_amd64.deb ...
Unpacking libipt2 (2.1.2-3) ...
Selecting previously unselected package libsource-highlight-common.
Preparing to unpack .../5-libsource-highlight-common_3.1.9-4.3build2_all.deb ...
Unpacking libsource-highlight-common (3.1.9-4.3build2) ...
Selecting previously unselected package libsource-highlight4t64:amd64.
Preparing to unpack .../6-libsource-highlight4t64_3.1.9-4.3build2_amd64.deb ...
Unpacking libsource-highlight4t64:amd64 (3.1.9-4.3build2) ...
Selecting previously unselected package gdb.
Preparing to unpack .../7-gdb_17.1-2ubuntu1_amd64.deb ...
Unpacking gdb (17.1-2ubuntu1) ...
Selecting previously unselected package libc6-dbg:amd64.
Preparing to unpack .../8-libc6-dbg_2.43-2ubuntu2.3_amd64.deb ...
Unpacking libc6-dbg:amd64 (2.43-2ubuntu2.3) ...
Selecting previously unselected package valgrind.
Preparing to unpack .../9-valgrind_1%3a3.26.0-0ubuntu1_amd64.deb ...
Unpacking valgrind (1:3.26.0-0ubuntu1) ...
Setting up libdebuginfod-common (0.194-4) ...
Setting up libsource-highlight-common (3.1.9-4.3build2) ...
Setting up libc6-dbg:amd64 (2.43-2ubuntu2.3) ...
Setting up libdw1t64:amd64 (0.194-4) ...
Setting up libipt2 (2.1.2-3) ...
Setting up libbabeltrace1:amd64 (1.5.11-5build1) ...
Setting up valgrind (1:3.26.0-0ubuntu1) ...
Setting up libdebuginfod1t64:amd64 (0.194-4) ...
Setting up libsource-highlight4t64:amd64 (3.1.9-4.3build2) ...
Setting up gdb (17.1-2ubuntu1) ...
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for libc-bin (2.43-2ubuntu2.3) ...
allur@AKSHAYA:~$ |
```

Step-5: Check that it installed

```
allur@AKSHAYA:~$ valgrind --version
valgrind-3.26.0
allur@AKSHAYA:~$ |
```

Step-7: Run your program with Valgrind

```
allur@AKSHAYA:~$ valgrind --leak-check=full ./a.out
==1446== Memcheck, a memory error detector
==1446== Copyright (C) 2002-2024, and GNU GPL'd, by Julian Seward et al.
==1446== Using Valgrind-3.26.0 and LibVEX; rerun with -h for copyright info
==1446== Command: ./a.out
==1446==
myshell> hello
myshell> ls
myshell> This is test command
myshell> exit

Commands:
This is test command
ls
hello
==1446==
==1446== HEAP SUMMARY:
==1446==   in use at exit: 0 bytes in 0 blocks
==1446==   total heap usage: 11 allocs, 11 frees, 2,196 bytes allocated
==1446==
==1446== All heap blocks were freed -- no leaks are possible
==1446==
==1446== For lists of detected and suppressed errors, rerun with: -s
==1446== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
allur@AKSHAYA:~$
```

Result

Command history, arrow-key navigation, dynamic buffer allocation, buffer resizing, and linked-list management were successfully implemented. Valgrind confirmed that all allocated memory was freed with zero errors and no memory leaks.